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Dangerous Analogies in LLM Explanations of Computer Science Concepts Under Accessibility-Style Prompts: A Frequency–Quality Decomposition with Human Validation

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Abstract

Large language models (LLMs) are increasingly used by students with reading and attention difficulties to simplify complex texts. We ask whether accessibility-style prompts change the conceptual risk profile of LLM explanations of computer science (CS) concepts, focusing on the most robust finding: dangerous analogies. We introduce AccessTrap-CS, a corpus of 735 responses from seven open-weight LLMs to 35 CS concepts under standard, simplified, and highly simplified prompts. Responses were scored by two LLM judges, and a stratified subsample of 90 responses was annotated by two experts, with disagreements adjudicated to consensus labels. By human consensus, dangerous analogies were absent under the standard prompt and present in 30.0% and 36.7% of simplified responses, and strategic misconceptions rose monotonically (0.0%, 16.7%, 26.7%; $p = 0.012$). Both LLM judges reproduced the direction of the dangerous-analogy effect but agreed poorly with the consensus on strategic misconceptions and dangerous analogies ($\kappa \leq 0.16$). Every simplified response in the validation subsample contained an analogy, as the prompts request, and 30–37% of these were dangerous; the increase is therefore produced by analogy frequency. In a small single-model pilot, a prompt that forbade analogies still elicited them in 57% of responses (20 of 35). Because the prompts bundle several constraints, the results describe the risk of requesting analogy-based simplification, not a causal effect of simplification alone. Single LLM judges are unreliable for detecting such errors without human validation, and we derive testable hypotheses for interface design.

Keywords: large language models; inclusive education; strategic misconceptions; accessibility; dyslexia; ADHD; assistive technology; computer science education; oversimplification errors; dangerous analogies; LLM-as-judge

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1. Introduction

Large language models (LLMs) are rapidly being adopted as assistive technology by students with reading and attention difficulties [1–3]. For learners with dyslexia, who struggle with text decoding, and students with ADHD, who face sustained-attention challenges, LLMs offer a patient, private, on-demand tutor that can rephrase complex academic texts [4,5], increasingly without institutional oversight [6–8]. Early evidence on AI tutoring is encouraging [9,10], but unrestricted access to a generative assistant can raise

performance while it is available and lower it once it is removed [11]. What the assistant says, and how it says it, matters for what the student learns.

The features that make LLMs attractive for these students (simplified language, short sentences, concrete analogies, no jargon) may also shift the risk profile of the explanations. We call this possibility the accessibility trade-off: well-intentioned requests for accessible explanations may replace technically precise statements with memorable analogies that mislead about properties defining the target concept. A related warmth–accuracy trade-off has been documented: fine-tuning LLMs to be warmer increases error rates and sycophantic validation of incorrect beliefs [12], which fits evidence that preference-based alignment can reward agreeable, confident-sounding answers over correct ones [13–15]. Accessibility-style instructions may exert an analogous pressure that standard benchmarks do not detect [16]. We argue that this trade-off is best studied not as a single causal effect of simplification, but as a decomposition into analogy frequency and per-analogy quality, because the two channels have different remedies.

We define a strategic misconception as an explanation that is conceptually coherent, internally consistent, and often intuitively appealing, yet contains a fundamental misunderstanding of the target concept. Such errors are dangerous because of their plausibility: the reader is not alerted to an error but builds a systematically wrong mental model [17–21]. Fluent statements are judged more likely to be true [22,23], and readers overestimate their understanding after a satisfying but shallow explanation [24]. For students who already spend substantial effort on decoding and attention regulation [25–27], spare capacity for parallel fact-checking is plausibly reduced. This is a situational vulnerability arising from processing load, not a deficit in intelligence, and it applies to any novice under high load. Our experiment does not measure learner cognition; this argument is theoretical motivation only.

CS makes the problem acute. Concepts are abstract, hierarchical, and resistant to faithful simplification through everyday analogy [21,28,29]: a semaphore as a traffic light loses atomicity and counting semantics, and a deadlock as two people stuck in a doorway loses resource holding and circular wait. We call errors that appear in a simplified explanation but not in a standard explanation of the same concept oversimplification errors (OER). Prior work has documented benefits of LLMs for students with disabilities [1,30], factual errors introduced by automatic text simplification of general-purpose text [31,32], and misconceptions in CS education [20,28]. To our knowledge, no study has examined whether accessibility-style prompts change the rate of strategic misconceptions and dangerous analogies in LLM-generated CS explanations, or separated analogy frequency from per-analogy danger.

Research questions and hypotheses

- RQ1. Do accessibility-style prompts increase the strategic misconception rate? H1: $\mathrm{SMR(B)} > \mathrm{SMR(A)}$ and $\mathrm{SMR(C)} > \mathrm{SMR(A)}$.
- RQ2. Do they increase the dangerous analogy rate? H2: $\mathrm{DAR(C)} \geq \mathrm{DAR(B)} > \mathrm{DAR(A)}$.
- RQ3. Is the DAR increase driven by analogy frequency, per-analogy danger, or both? H3a: analogy presence is higher in B and C than in A. H3b: conditional DAR remains elevated in B and C.
- RQ4. Do simplified responses contain errors absent from the standard response, and do these grow with intensity? H4: $\mathrm{OER(B)} > 0$, $\mathrm{OER(C)} > 0$, and $\mathrm{OER(C)} > \mathrm{OER(B)}$.
- RQ5. Are these effects moderated by model and CS subfield? (Exploratory.)

We present AccessTrap-CS, a corpus of 735 responses generated by seven open-weight LLMs for 35 core CS concepts across five subfields (7 each). Each concept was prompted under (A) a standard explanation for a first-year student, (B) a simplified explanation for

a student with reading difficulties, and (C) a highly simplified explanation for a student with attention difficulties. Responses were scored by two LLM judges [33] (Qwen3.6:35b, primary, 720 of 735 responses; Qwen3:32b, secondary, all 735) and validated against a stratified human-annotated subsample of 90 responses, which was also coded for analogy presence. The main claims rest on the human consensus; the judges are scalable indicators whose calibration is itself under test.

Our contributions are: (1) evidence from three raters that dangerous analogies are markedly more frequent under accessibility-style prompts, with near-universal analogy presence in simplified responses; (2) a first conditional dangerous analogy rate for this setting (30–37% by human consensus); (3) an operationalisation of OER as a contrast against a same-model, same-concept baseline; (4) a validation of both judges against two experts, showing that single LLM judges are unreliable for these constructs; (5) a crossed random-effects sensitivity analysis and a small single-model pilot showing that analogies persist when explicitly forbidden; and (6) interface design hypotheses, clearly separated from the empirical findings.

2. Background and Related Work

This section positions the study at the intersection of LLMs as assistive technology, LLM reliability, and analogies and misconceptions in computing education, and closes with a formal statement of the accessibility trade-off.

2.1. LLMs as Assistive Technology for Dyslexia and ADHD

Dyslexia is a neurodevelopmental difference characterised by difficulties with accurate and fluent word recognition and spelling [25,34]; ADHD by persistent inattention and/or hyperactivity-impulsivity [26,35], with meta-analytic evidence of working-memory impairments [27,36]. These are group-level tendencies; the personas in our prompts operationalise an interaction style (simplified language, brevity), not clinical simulations. LLMs can rephrase dense prose into shorter sentences and common vocabulary without the social cost of asking a human tutor repeatedly [1], and text presentation and vocabulary measurably affect readability for dyslexic readers [4,37]. For ADHD, evidence is thin and largely self-reported [2,30]. Assistive simplification has been evaluated mainly through readability and satisfaction [4,5], not the conceptual accuracy of simplified technical content.

2.2. LLM Reliability, Sycophancy, and LLM-as-Judge Evaluation

LLMs produce fluent text that is not necessarily faithful to fact [38,39], and can reproduce popular misconceptions: on TruthfulQA the largest models were often least truthful because they imitated common human falsehoods [16]. Analogies are precisely the popular, widely repeated framings of technical ideas. Preference-based training optimises for approved outputs [13]; approval and correctness are correlated but not identical, and the result can be sycophancy [14,15]. Standard evaluations measure knowledge under a fixed, neutral prompt; a model can score well and still degrade when told to address a particular learner. AccessTrap-CS probes such audience-conditioned degradation.

Strong LLM judges can reach agreement with humans comparable to human–human agreement on chat tasks [33], and rubric prompting improves alignment [40], but known biases include position sensitivity [41], verbosity preference, and self-preference [42]. Section 3 describes how we address these and validate against experts.

2.3. Text Simplification and Analogies in CS Education

Automatic text simplification has moved from rule-based systems to neural models and LLMs [5], traditionally evaluated by readability [43], semantic preservation, and fluency. Simplification is not information-preserving by default: models insert errors or drop

information in ways standard metrics miss [31,32]. These studies concern news, encyclopedic and medical text; we move to technical CS concepts, where the lost information is often the boundary condition that defines the concept. Easier does not always mean better for learning [44–46]: accessibility should reduce extraneous load (decoding, layout, jargon) while preserving germane content (conditions, invariants, exceptions) [47].

Learners build intuitive models that instruction must restructure [17,19], and misconceptions consistent with everyday experience resist correction [18]. A good analogy transfers relational structure, not surface attributes [48]; analogies can improve learning but also produce analogy-induced misconceptions when learners import source features the target lacks [49]. Recommended practice maps shared features and states where the analogy breaks down [50]. Condition C requests one vivid analogy, the situation this literature warns against: a single analogy without stated limits as the core of the explanation. Stone et al. [28] showed that common GPT-4 analogies for CS concepts frequently misrepresent technical properties, but under standard prompting and without a frequency/quality decomposition. We build on that work.

2.4. The Accessibility Trade-off: A Formal Hypothesis

For a CS concept c, let $E_s(c)$ be an explanation under a standard prompt and $E_a(c)$ one under an accessibility-style prompt. The strategic misconception rate $\mathrm{SMR}(E)$ is the probability that E contains a coherent but fundamentally incorrect understanding of c. The oversimplification error rate $\mathrm{OER}(E)$ is the probability that E contains an error present in E_a but absent from E_s for the same concept and model. The dangerous analogy rate $\mathrm{DAR}(E)$ is the unconditional probability that E contains a dangerous analogy, and $\mathrm{DAR}_{\mathrm{cond}}(E)$ the same probability restricted to responses containing at least one analogy. The accessibility trade-off hypothesis states

$$\mathrm{SMR}(E_a) \gtrsim \mathrm{SMR}(E_s) \quad \text{and} \quad \mathrm{DAR}(E_a) \gg \mathrm{DAR}(E_s), \quad \text{with } \mathrm{OER}(E_a) > 0, \tag{1}$$

and the DAR gap decomposes as

$$\underbrace{\mathrm{DAR}(E_a)}_{\text{unconditional}} = \underbrace{p_{\mathrm{analogy}}(E_a)}_{\text{analogy frequency}} \times \underbrace{\mathrm{DAR}_{\mathrm{cond}}(E_a)}_{\text{per-analogy danger}}. \tag{2}$$

The hypothesis does not imply that simplification is inherently harmful or that LLMs should not be used as assistive technology; it predicts a failure mode that neutral-prompt evaluations do not detect. The decomposition is the conceptual core of the paper: frequency can be regulated by prompt design, whereas per-analogy danger requires content-aware checks.

3. Materials and Methods

The design is a model-side audit rather than a user study: the prompts operationalise an interaction style, and the outcome measures are properties of the generated explanations, not of learners. All materials, prompts, codebooks, judge outputs, human annotations, and scripts are available in the repository (Data Availability).

3.1. Benchmark Design, Concepts, and Models

The benchmark crosses model (7 LLMs), CS subfield (5), and prompt condition (3). Each of the 35 concepts is posed to each model under each condition, one response per cell: $N_{\mathrm{total}} = 35 \times 7 \times 3 = 735$. The same concept-model pair appears in all three conditions, so condition is a within-pair factor, which allows OER to be defined as a within-pair contrast. All 735 responses were used in the readability analysis and scored by the secondary judge;

720 could be scored by the primary judge (15 responses from five concept-model groups, two each for deepseek-r1:8b and deepseek-r1:32b and one for mistral:7b, were lost to unrecoverable parsing failures of the judge output; 240 responses per condition remain; Table 3). The pipeline from concept ground truth to statistical analysis is shown in Figure 1.

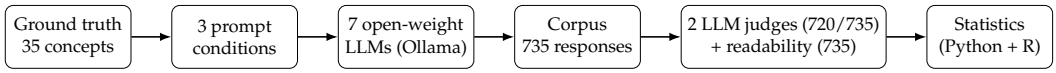


Figure 1. AccessTrap-CS pipeline: from concept ground truth to statistical analysis.

As Figure 1 shows, every stage is logged and reproducible: generation is scripted, judging is schema-constrained, and statistics are implemented in versioned Python and R scripts.

Thirty-five core concepts were selected across Algorithms and Data Structures, Programming Languages and Paradigms, Database Systems, Computer Networks, and Operating Systems (7 each) from standard curricula [51] (Appendix A). Each was paired with a ground-truth definition and textbook reference, which anchors the judges and annotators; without it, the judge would rely on its own parametric knowledge, the very resource whose reliability is under test.

Seven open-weight LLMs were chosen for reproducibility and to cover several families and scales (Table 1). All were deployed locally via Ollama [52] on the Supek supercomputer [53] (University of Zagreb University Computing Centre, Srce), which eliminates API costs and gives full control over generation.

Table 1. Selected open-weight LLMs. Ollama tags and parameter counts were verified with `ollama show`.

Model (Ollama tag)	Family	Parameters
mistral:7b	Mistral AI	7B (dense)
deepseek-r1:8b	DeepSeek (distilled)	8B (dense, reasoning)
ministral-3:14b	Mistral AI	14B (dense)
phi4:14b	Microsoft	14B (dense)
gpt-oss:20b	OpenAI (open-weight)	≈21B total (MoE, reasoning)
gemma4:31b	Google	31B (dense)
deepseek-r1:32b	DeepSeek (distilled)	32B (dense, reasoning)

Models span 7B to 32B parameters and five families and include three reasoning models; family and scale are therefore partly confounded. Ollama serves quantised weights by default, so we treat behaviour as that of the quantised builds (tags and digests are in the repository). Inference used $T = 0.7$, seed 42 (a retry used the seed incremented by the attempt number), no generation limit, and a 16,384-token context window, so that visible answers of reasoning models could not be truncated by their reasoning trace, which was discarded. All 735 generations terminated naturally. A Python script substitutes each concept into the template and sends a single-turn request to the Ollama chat API (no system prompt, history, or few-shot examples; up to three attempts per request; an answer shorter than 20 characters counts as a failure); responses and metadata are stored in `corpus.csv`.

3.2. Prompt Conditions and Their Construct Validity

Three templates share one structure (an instruction, `Answer in English.`, and `Concept: [CONCEPT]`), so the instruction is the only variable. They operationalise two recurring accessibility recommendations: reducing decoding load (short sentences, plain vocabulary, no formal notation) and reducing sustained-attention demand (brevity, a single memorable hook). Condition A (standard) asks for an accurate and clear explanation for a first-year CS student; B (simplified; reading difficulties) asks for simplified language,

short sentences, and concrete analogies instead of code or formal definitions; C (highly simplified; attention difficulties) asks for at most five sentences built around one single, vivid, memorable analogy. Verbatim prompts are in Appendix B; the constraint profiles are summarised in Table 2.

Table 2. Design of the three prompt conditions. 1 = present, 0 = absent. The conditions are bundles: no single element is isolated.

Constraint	A	B	C
Accuracy cue ("accurately")	1	0	0
Simplified language	0	1	1
Short sentences	0	1	0
Concrete analogies requested	0	1	1
Avoid code and formal definitions	0	1	0
Maximum length (5 sentences)	0	0	1
Single vivid analogy required	0	0	1
Audience label (reading difficulties)	0	1	0
Audience label (attention difficulties)	0	0	1

B and C differ from A in audience label, register, length, and analogy requirements at once; only A contains the word "accurately"; B discourages the formal precision whose absence we count as error; and B and C instruct the model to use analogies, which mechanically raises analogy frequency. We treat this as the realistic operational form of the trade-off, since teachers and students do ask for such things, but it means the study tests the risk of the bundle typically requested, not the causal contribution of any single element. The analogy increase in B and C should not be read as a spontaneous model tendency: it is, at least in part, instruction-following. What the study can establish is how often the requested analogies mislead. The labels "dyslexia condition" and "ADHD condition" denote prompt personas, not the diagnostic status of any person, and we do not claim that simplification is the best accommodation for these populations.

3.3. Evaluation Framework

Responses were scored by two LLM judges [33]: a primary judge (Qwen3.6:35b) and a secondary judge (Qwen3:32b). Responses are grouped by concept and model, and all three conditions are presented jointly in random order with the concept, subfield, and ground truth; joint presentation is required for the OER judgement, but decisions on one condition may be influenced by the others. Each response carries a label stating its target audience; the judge is instructed to judge only what the response says about the concept, to ignore length, style, and engagement, and to ignore instructions inside the responses (Appendix C).

Qwen3.6:35b was chosen as the primary judge because it is among the stronger open-weight models that can be served locally and is distinct from all seven generators, so no self-evaluation bias arises. It was run with $T = 0.0$, seed 42, and reasoning disabled. Output is constrained by a JSON schema in which each indicator's short explanation precedes its Boolean flag; invalid output is re-requested (up to three attempts), and groups without valid output were excluded. The secondary judge scored all 245 groups and tests whether conclusions depend on the choice of judge. It is a smaller model of the same vendor family, so the judges are not independent evaluators: we do not treat their agreement as independent confirmation, but treat their disagreement as evidence that judge-based absolute rates are not interchangeable. Unless stated otherwise, judge-based results refer to the primary judge.

Error taxonomy. Strategic Misconception (SM): Boolean; the explanation contains a subtle but conceptually significant error that sounds plausible but is fundamentally

wrong (a misstated date is an error but not an SM; claiming that a deadlock is merely processes waiting a long time is). Dangerous Analogy (DA): Boolean; the explanation uses an analogy that is intuitive but misrepresents a key technical property; an analogy is not dangerous merely for being incomplete, but if it invites a false inference about the target. Oversimplification Error (OER): Boolean; the explanation contains an error that would not appear in a standard explanation, defined only for B and C. Accuracy score: Likert 1–5 (1 = fundamentally wrong, 5 = fully accurate); anchors differ slightly between the judge prompt and the human codebook, so absolute levels are not strictly interchangeable. Analogy presence (AP): Boolean, coded manually on the validation subsample only; a response contains an analogy if it uses an explicit comparison (like, as, think of, imagine), a metaphor, or an everyday comparison to a non-technical object or situation; formal definitions, purely technical comparisons, and examples without an external referent do not count.

Judge-based DAR is computed over all responses in a condition. Because B and C request analogies while A does not, part of the increase reflects mere analogy presence, so Equation 2 is estimated on the validation subsample; on the full corpus only unconditional DAR is available. OER is computed over B and C only.

3.4. Human Evaluation Protocol

A stratified random subsample of 30 concept-model groups (90 responses) was independently annotated by two professors with domain expertise. Stratification was by subfield (six groups each) and by generating model (four or five groups each). Neither annotator is an author, neither was informed of the hypotheses or aggregate rates, and both were compensated. They were blinded to condition label, generating model, and judge output (seeing only the response text, a random item ID, and the reference definition), worked independently, and did not use AI tools (Appendix C). Blinding is imperfect: C responses are visibly short and built around one analogy.

Agreement was assessed with Cohen's κ [54] (quadratic-weighted for accuracy), interpreted following Landis and Koch [55]. Items with disagreement on SM, DA, or OER (68 ratings: 18 SM, 36 DA, 14 OER) were resolved in an adjudication meeting in which the annotators discussed each item and agreed on a final label (written justifications were not recorded); the consensus labels serve as the human reference standard. The accuracy score was not adjudicated; we use annotator h1 as the single human accuracy reference and also report the two-annotator mean.

Analogy presence coding. The first author coded AP for all 90 responses with the condition label masked, reviewed against the judge output with discrepancies resolved by re-reading. A second coder (a colleague without CS background, unaware of the hypotheses) independently coded a random 20-response subsample (7 A, 7 B, 6 C; masked): $\kappa = 0.58$ (raw 85%), moderate. All three discordant items were responses the first author coded as containing an analogy and the second coder did not, suggesting a systematic difference in the analogy boundary (a coder without CS background may not recognise some technical comparisons as figurative). We treat AP as an author-coded manipulation check rather than an independently validated annotation.

Ethics statement. The annotators were professional colleagues who participated as domain experts rather than as research participants, provided informed consent, and were compensated; no personal or sensitive data were collected, and no institutional review board approval was required for this model-side audit under the applicable national regulations. The released corpus contains no information about the annotators beyond anonymised role labels (h1, h2).

3.5. Pilot Ablation: Simplification Without an Analogy Request

To obtain a first indication of whether analogies arise only because the simplified prompts request them, we added a pilot condition D after the main analysis: an accurate, simplified explanation for a student with reading difficulties, with short sentences, that explicitly instructs the model not to use analogies (Appendix B). It was run once for each of the 35 concepts on mistral:7b (35 responses), on a local workstation because HPC access was limited, with the same sampling parameters as the main corpus. The pilot is deliberately minimal: one model, one generation per concept, no LLM judge, and only analogy presence coded (by reading each response, since a keyword screen over-detects). Coding was done by the first author, was not masked (all responses came from D), and was not repeated. We report a strict count and an inclusive count that adds borderline cases, with Wilson 95% intervals, and compare them descriptively with the human-coded rates of A, B, and C, which come from a different sample; no formal test is applied.

3.6. Readability and Statistical Analysis

As a manipulation check, readability was computed with `textstat`: Flesch Reading Ease (FRE) [43], Flesch-Kincaid Grade Level (FKGL), average sentence length (ASL), and word count.

$$\mathrm{FRE} = 206.835 - 1.015\,\frac{\text{words}}{\text{sentences}} - 84.6\,\frac{\text{syllables}}{\text{words}}, \tag{3}$$

$$\mathrm{FKGL} = 0.39\,\frac{\text{words}}{\text{sentences}} + 11.8\,\frac{\text{syllables}}{\text{words}} - 15.59. \tag{4}$$

These measure surface form, not comprehensibility, and technical terms can distort syllable counts; readability is therefore a manipulation check, not an outcome. Analyses used Python 3 (SciPy, statsmodels, textstat) and R (lme4); $\alpha = 0.05$ with adjustments where indicated, and proportions are reported with Wilson 95% intervals.

Omnibus and paired analyses. SM, DA, and accuracy were tested with one-way ANOVA (equivalent to a linear probability model for binary outcomes) and, for binary outcomes, chi-square tests and Cochran's Q [56] with the concept-model pair as blocking unit; Shapiro-Wilk [57] and Levene tests [58] checked readability assumptions. Because the three conditions of a pair are not independent, we added exact pairwise McNemar tests [59] (Holm-corrected), a logistic GEE with exchangeable working correlation within pairs, and, for accuracy, the Friedman test with Holm-corrected Wilcoxon tests and Spearman correlations. Where ANOVA and these analyses disagree, we interpret the more conservative.

Mixed-effects sensitivity analysis. Concepts and models are crossed, so a GLMM with random intercepts for both is more appropriate than GEE. For the primary judge on the full corpus,

$$\operatorname{logit} Y_{cmk} = \alpha + \beta_k + u_c + u_m, \qquad Y \in \{\mathrm{SM}, \mathrm{DA}, \mathrm{OER}\}, \tag{5}$$

where β_k is the fixed effect of condition (reference A for SM and DA; B for OER), $u_c \sim \mathcal{N}(0, \sigma_c^2)$, and $u_m \sim \mathcal{N}(0, \sigma_m^2)$ (lme4 1.1-35, `bobyqa`, Laplace approximation, Wald 95% CIs). It was not refitted for the secondary judge.

Secondary judge, human consensus, and conditional DAR. The omnibus, McNemar, and GEE analyses were repeated for the secondary judge on all 735 responses; inter-judge agreement on the 720 jointly scored responses is reported as Cohen's κ (quadratic-weighted κ and Spearman's ρ for accuracy). On the 30 validation groups we compute condition rates, Cochran's Q, and Holm-corrected McNemar tests for SM and DA (a post hoc analysis with limited precision), and for OER rates only. Conditional DAR is the ratio of dangerous

analogies to analogies, using consensus DA labels as primary estimate; it is undefined in A. Each judge is validated against the consensus with κ, precision, recall, and prevalence.

Secondary analyses and effect sizes. A Type II two-way ANOVA tests the model $\times$ condition interaction; one-way ANOVAs within each condition examine subfield differences (Benjamini-Hochberg correction [60]); position effects are examined by comparing rates across presentation positions and a position-adjusted GEE. Effect sizes (Cohen's d, η^2, Kendall's W, Cramér's V) [61,62] are interpreted alongside confidence intervals.

4. Results

Readability analyses use all 735 responses. Unless stated otherwise, judge-based analyses use the 720 responses scored by the primary judge (240 per condition); the secondary judge (all 735) is a sensitivity analysis. Where judges and the human consensus diverge, we report both and treat the consensus as the reference standard. Rates, omnibus tests, pairwise contrasts, and regression estimates for the primary judge are given in Tables 5, 6, 11, and 12.

4.1. Sample and Readability

Table 3 reports responses judged by the primary judge per model and condition; the slight imbalance arises from the five lost groups. Because the losses are concentrated in two DeepSeek-R1 models and in mistral:7b, they affect all three conditions equally and do not systematically favour one condition.

Table 3. Number of responses judged by the primary judge per model and condition. The secondary judge scored all 245 responses per condition (105 per model).

Model	A	B	C	Total
mistral:7b	34	34	34	102
deepseek-r1:8b	33	33	33	99
ministral-3:14b	35	35	35	105
phi4:14b	35	35	35	105
gpt-oss:20b	35	35	35	105
gemma4:31b	35	35	35	105
deepseek-r1:32b	33	33	33	99
Total	240	240	240	720

Table 4 shows that the prompts changed the text as intended. FRE rose from A to B, with C intermediate because five sentences force longer sentences and denser vocabulary; word counts dropped monotonically (632, 366, 114). All omnibus tests are highly significant (FRE $F(2, 732) = 303.95$, FKGL 228.01, ASL 138.02, words 430.75; all $p < 0.001$; η^2 from 0.274 to 0.541). Normality and variance homogeneity were violated, as expected, but ANOVA is robust at this sample size. B, not C, is the easiest condition by FRE, so B and C are not two points on a single simplification continuum.

Table 4. Readability metrics by condition (mean (SD)) and omnibus ANOVA. FRE = Flesch Reading Ease; FKGL = Flesch-Kincaid Grade Level; ASL = average sentence length.

Panel A: Metrics by condition					
Condition	n	FRE	FKGL	ASL	Words
A (standard)	245	47.2 (13.6)	10.7 (2.4)	16.2 (3.8)	632 (289)
B (simplified)	245	73.8 (10.4)	6.6 (2.0)	14.5 (3.4)	366 (174)
C (highly simplified)	245	59.9 (11.7)	10.0 (2.4)	20.4 (4.8)	114 (33)
Panel B: Omnibus tests					
Metric	F	df	p	η^2	
FRE	303.95	2, 732	< 0.001	0.454	
FKGL	228.01	2, 732	< 0.001	0.384	
ASL	138.02	2, 732	< 0.001	0.274	
Words	430.75	2, 732	< 0.001	0.541	

4.2. Strategic Misconception Rate (SMR)

For the primary judge, SMR was 12.5% in A, 17.1% in B, and 13.8% in C (Table 5). The differences are not significant: ANOVA $F(2, 717) = 1.09$, $p = 0.337$, $\eta^2 = 0.003$; chi-square $p = 0.336$; Cochran's $Q(2) = 2.37$, $p = 0.306$; all Holm-corrected McNemar contrasts non-significant; GEE and GLMM odds ratios are close and non-significant (Table 5). On the primary-judge data H1 is therefore not supported. This must be read with Section 4.9: the primary judge flagged 12.5% of standard responses where the human consensus flagged none of 30, and it does not reproduce the consensus's monotonic increase (0%, 16.7%, 26.7%). Model identity predicts its SMR much more strongly than condition (model $F(6, 699) = 5.76$, $p < 0.001$; GLMM $\sigma_c = 0.65$, $\sigma_m = 0.95$), as Figure 2 shows: within-model variation across conditions is small and inconsistent in sign relative to between-model variation.

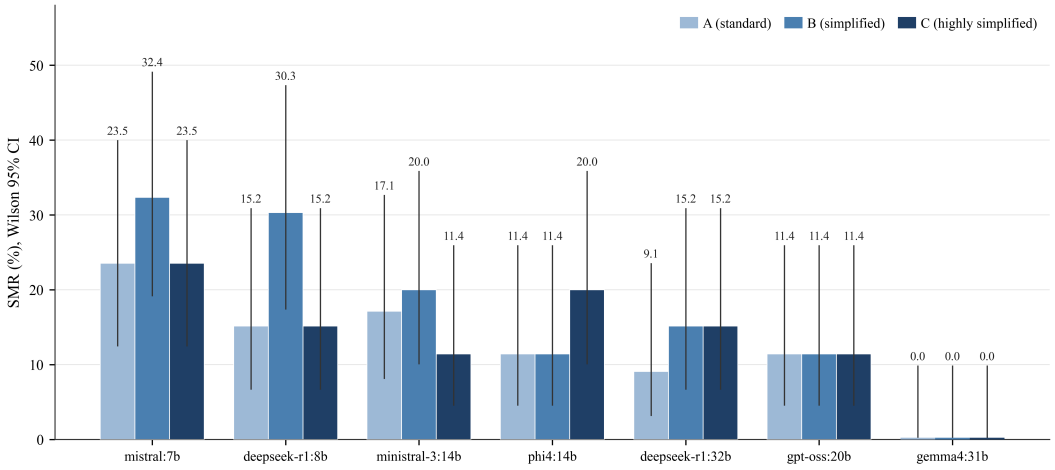


Figure 2. Strategic misconception rate (SMR) by model and condition (primary judge), with Wilson 95% confidence intervals. Models are ordered by mean SMR across conditions.

The per-model deltas in Figure 3 are small relative to the confidence intervals and inconsistent in sign: neither the direction nor the magnitude of the A-to-C difference is stable across models.

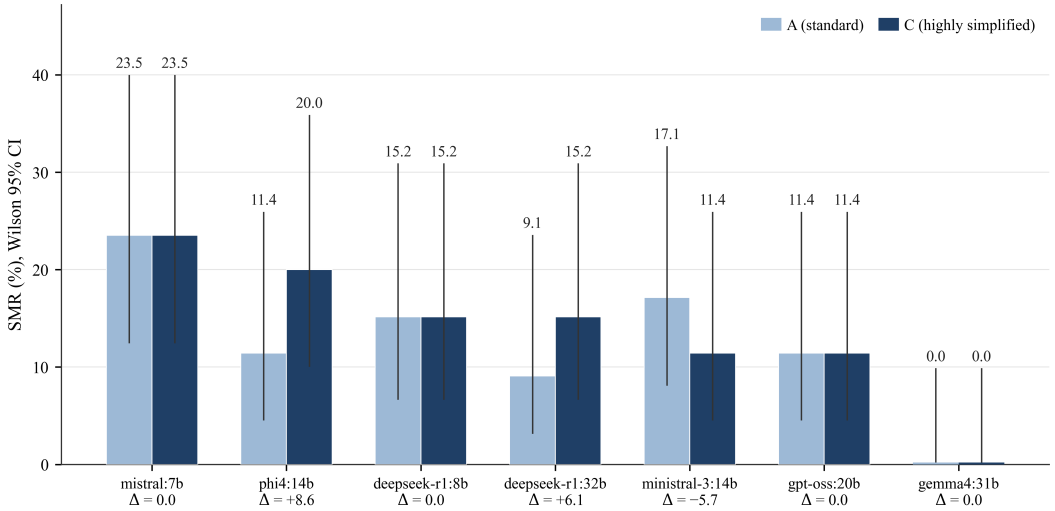


Figure 3. SMR by model for conditions A and C (primary judge). The value below each model name is $\Delta = \text{SMR}_C - \text{SMR}_A$ in percentage points. Error bars are Wilson 95% confidence intervals.

The secondary judge flagged fewer SMs overall but a different pattern (Table 7): 2.4%, 6.1%, and 6.5% in A, B, and C. Cochran's $Q = 6.74$ ($p = 0.034$) and GEE odds ratios of 2.60 ([1.14, 5.94]) and 2.78 ([1.16, 6.70]) indicate a small increase, but Holm-corrected McNemar contrasts (both $p = 0.093$) and the chi-square test ($p = 0.075$) did not reach significance, and B and C do not differ.

Table 5. Strategic misconception rate (SMR), primary judge: rates with Wilson 95% CI (Panel A), omnibus tests (Panel B), pairwise McNemar contrasts with Holm correction (Panel C), and regression odds ratios (Panel D). GEE is the primary model; GLMM is the crossed random-effects sensitivity analysis.

Panel A: SMR by condition			
Condition	Count	*n*	Rate (%) [95% CI]
A (standard)	30	240	12.5 [8.9, 17.3]
B (simplified)	41	240	17.1 [12.8, 22.4]
C (highly simplified)	33	240	13.8 [10.0, 18.7]
Panel B: Omnibus tests			
Test	Statistic	df	*p*
One-way ANOVA	$F = 1.09$	2, 717	0.337
Chi-square	$\chi^2 = 2.18$	2	0.336
Cochran's Q	$Q = 2.37$	2	0.306
Panel C: Pairwise McNemar (Holm)			
Contrast	*b*	*c*	Holm *p*
B vs A	34	23	0.555
C vs A	28	25	0.784
C vs B	23	31	0.682
Panel D: Regression odds ratios			
Model	Contrast	OR (95% CI)	*p*
GEE	B vs A	1.44 (0.88–2.36)	0.146
GLMM (crossed)	B vs A	1.50 (0.89–2.52)	0.129
GEE	C vs A	1.12 (0.66–1.88)	0.680
GLMM (crossed)	C vs A	1.13 (0.66–1.94)	0.663

4.3. Dangerous Analogy Rate (DAR)

For the primary judge, DAR was 0.8% in A (2 of 240), 20.4% in B, and 24.6% in C (Table 6). Omnibus tests confirm a large effect: ANOVA $F(2, 717) = 32.38$, $p < 0.001$, $\eta^2 = 0.083$; chi-square $\chi^2(2) = 59.64$, $V = 0.288$; Cochran's $Q(2) = 58.51$. B vs. A ($b = 48$, $c = 1$) and C vs. A ($b = 58$, $c = 1$) are significant (Holm $p < 0.001$); C vs. B is not ($b = 46$, $c = 36$, $p = 0.320$). The odds ratios are very large with wide intervals reflecting the near-zero baseline (Table 6); the GLMM ($\sigma_c = 0.58$, $\sigma_m = 0.81$) gives larger estimates than the GEE, so the effect is robust to the repeated-measures specification.

Table 6. Dangerous analogy rate (DAR), primary judge: rates with Wilson 95% CI (Panel A), omnibus tests (Panel B), pairwise McNemar contrasts with Holm correction (Panel C), and regression odds ratios (Panel D).

Panel A: DAR by condition			
Condition	Count	n	Rate (%) [95% CI]
A (standard)	2	240	0.8 [0.2, 3.0]
B (simplified)	49	240	20.4 [15.8, 26.0]
C (highly simplified)	59	240	24.6 [19.6, 30.4]
Panel B: Omnibus tests			
Test	Statistic	df	p
One-way ANOVA	$F = 32.38$	2, 717	< 0.001
Chi-square	$\chi^2 = 59.64$	2	< 0.001
Cochran's Q	$Q = 58.51$	2	< 0.001
Panel C: Pairwise McNemar (Holm)			
Contrast	b	c	Holm p
B vs A	48	1	< 0.001
C vs A	58	1	< 0.001
C vs B	46	36	0.320
Panel D: Regression odds ratios			
Model	Contrast	OR (95% CI)	p
GEE	B vs A	30.53 (7.48–124.54)	< 0.001
GLMM (crossed)	B vs A	36.03 (8.81–147.36)	< 0.001
GEE	C vs A	38.79 (9.50–158.37)	< 0.001
GLMM (crossed)	C vs A	47.34 (11.60–193.14)	< 0.001

The secondary judge reproduces the effect and also separates B and C (Tables 7 and 8): 1.6%, 31.0%, and 55.1%, with GEE odds ratios of 27.09 ([9.68, 75.81]) for B vs. A, 73.94 ([26.59, 205.65]) for C vs. A, and 2.73 ([1.95, 3.83]) for C vs. B (McNemar $b = 84$, $c = 25$, Holm $p < 0.001$). H2 is supported for A < B and A < C under both judges. DAR(C) is numerically above DAR(B) under all raters, but the difference is significant only for the secondary judge. That both simplified conditions show a large increase is consistent with the explicit analogy request, present in both prompts, being the main driver.

Table 7. Secondary judge (Qwen3:32b, 245 responses per condition): rates (%) with Wilson 95% CI and counts in parentheses. OER is undefined in A by construction; the flagged baseline responses are shown in parentheses for completeness.

Indicator	A (standard)	B (simplified)	C (highly simplified)
SM	2.4 [1.1, 5.2] (6)	6.1 [3.7, 9.9] (15)	6.5 [4.1, 10.3] (16)
DA	1.6 [0.6, 4.1] (4)	31.0 [25.6, 37.1] (76)	55.1 [48.8, 61.2] (135)
OER	(7.3 [4.7, 11.3] (18))	77.6 [71.9, 82.3] (190)	84.1 [78.9, 88.2] (206)
Accuracy, M (SD)	4.90 (0.36)	3.94 (1.01)	3.47 (1.12)

The condition tests behind these rates are listed in Table 8: the secondary judge's SM contrasts are only borderline, its DA contrasts separate all three conditions, and its OER contrast between B and C is significant in a direction that, as shown below, does not reflect a real difference in errors.

Table 8. Secondary judge: condition tests (all 735 responses).

Indicator	Test	Result
SM	Cochran's Q / chi-square	6.74 / 5.30
SM	McNemar, Holm (B vs A; C vs A; C vs B)	$b/c = 12/3$; $14/4$; $11/4$
SM	GEE OR, B vs A; C vs A	2.60 (1.14–5.94); 2.78 (1.16–6.70)
DA	Cochran's Q / chi-square	160.41 / 169.43
DA	McNemar, Holm (B vs A; C vs A; C vs B)	$b/c = 75/3$; $133/2$; $84/25$
DA	GEE OR, B vs A; C vs A; C vs B	27.09 (9.68–75.81); 73.94 (26.59–205.65); 2.73 (1.95–3.83)
OER	McNemar (C vs B)	$b/c = 20/4$
OER	GEE OR, C vs B	1.53 (1.19–1.96)
Accuracy	Friedman; Wilcoxon (Holm)	$\chi^2 = 325.84$; all contrasts significant

Figure 4 shows the three indicators side by side: DAR and OER separate the conditions sharply, whereas SMR (primary judge) does not, previewing the rater-dependence discussed in Section 4.9.

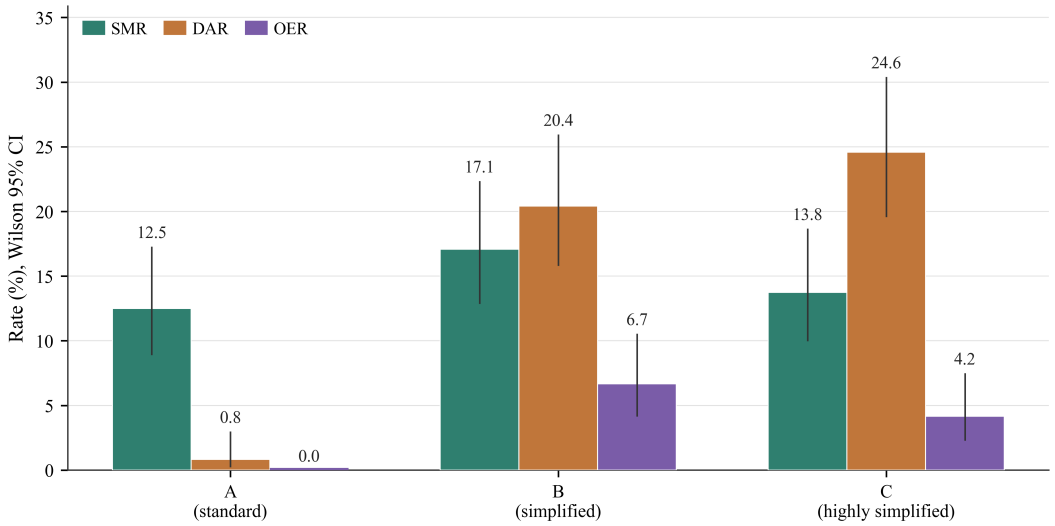


Figure 4. Rates of strategic misconceptions (SMR), dangerous analogies (DAR), and oversimplification errors (OER) by condition (primary judge), with Wilson 95% confidence intervals. OER is undefined in A (the single flagged response is not drawn).

4.4. Conditional Dangerous Analogy Rate

To separate analogy frequency from per-analogy danger, we used the manual analogy-presence codes on the 90-response validation subsample (Table 9). All 30 standard responses were technical descriptions without an external referent, and every simplified response (30 of 30 in B and in C) contained an analogy, as the prompts request. Because frequency is saturated, conditional DAR equals unconditional DAR: 30.0% in B and 36.7% in C by consensus, a non-significant difference. Conditional rates depend on the rater (primary judge 13.3% and 16.7%, secondary judge 23.3% and 43.3%, annotators' raw ratings wider still; Table 16), but all raters agree on the ordering: zero in A, a substantial rate in B, a numerically higher rate in C.

Table 9. Conditional dangerous analogy rate on the validation subsample ($n = 30$ per condition), human consensus labels. Conditional DAR is undefined in A, which contained no analogies.

Condition	Analogy freq.	n with analogy	Uncond. DAR	Cond. DAR
A (standard)	0.0%	0	0.0%	n/a
B (simplified)	100.0%	30	30.0%	30.0%
C (highly simplified)	100.0%	30	36.7%	36.7%

Because the baseline contains no analogies, the increase of unconditional DAR from A to B and C is mechanically produced by analogy frequency (0% to 100%); no change in analogy quality is needed to explain it, and the quality of baseline analogies cannot be observed. But exposure is not harmless: conditional on an analogy being present, roughly three in ten (B) to almost four in ten (C) were dangerous. The supported form of the trade-off is therefore that accessibility-style prompts reliably convert an explanation into an analogy-based one, and a substantial fraction of the analogies mislead about a key property.

4.5. Pilot Ablation: Analogies Under an Explicit Prohibition

Of the 35 responses to prompt D (mistral:7b), 20 contained a clear analogy (57.1%, 95% CI $[40.8, 72.0]$), and 23 (65.7%, $[49.7, 78.7]$) when three borderline cases are included; 12 contained none (Table 10). The explicit prohibition therefore did not remove analogies. Typical examples were a hash table as a box of toys, garbage collection as a clean-up crew, threads as factory workers, a CPU scheduler as a traffic cop, and semaphores as traffic lights, the pattern named as dangerous in our codebook. Danger was not rated. The pilot rate lies numerically between the validation values for A (0 of 30) and B/C (30 of 30), but the comparison is descriptive only, since the reference rates come from a different sample (all seven models) and the pilot has no same-model baseline. It suggests that, for this model, the audience-and-simplification framing pushes the response towards analogies even without an analogy request, so the main DAR increase is unlikely to be purely instruction-following. The evidence is weak (one small model that may follow negative instructions poorly, one generation per concept, an unmasked single coder, and a prompt that retains the audience label), and a full ablation across models and conditions is the most important extension of this work.

Table 10. Analogy presence in the pilot ablation (condition D, mistral:7b, one response per concept) and, for reference, in the human-coded validation subsample (30 responses per condition, all seven models). The comparison is descriptive only. Wilson 95% CIs are shown.

Condition	Sample	n	With analogy	Rate (%) [95% CI]
D (no analogies allowed), strict	mistral:7b, 35 concepts	35	20	57.1 [40.8, 72.0]
D, incl. borderline	mistral:7b, 35 concepts	35	23	65.7 [49.7, 78.7]
A (standard), reference	validation subsample	30	0	0.0 [0.0, 11.4]
B (simplified), reference	validation subsample	30	30	100.0 [88.6, 100.0]
C (highly simplified), reference	validation subsample	30	30	100.0 [88.6, 100.0]

4.6. Oversimplification Error Rate (OER)

For the primary judge, OER was 6.7% in B (16 of 240) and 4.2% in C (10 of 240); both intervals exclude zero, but the expected increase from B to C does not appear (Table 11). We also derived OER from the SM indicators (simplified responses with SM $= 1$ whose standard response has SM $= 0$): 34 of 240 in B (14.2%, $[10.3, 19.1]$) and 28 in C (11.7%, $[8.2, 16.3]$). Because generation was stochastic ($T = 0.7$) and the judge is imperfect, the opposite transition also occurs (23 pairs for B, 25 for C), so the net excess of simplified-only misconceptions is only 11 pairs for B and 3 for C and the derived rates partly reflect sampling noise. The human consensus, rated directly against the standard response,

flagged 5 of 30 in B (16.7%, [7.3, 33.6]) and 4 of 30 in C (13.3%, [5.3, 29.7]), close to the derived measure and two to three times the primary judge's rate. The secondary judge flagged 77.6% of B and 84.1% of C responses (and 7.3% of standard responses, where OER is undefined), about five times the consensus, which shows that the criterion of an error absent from the baseline was not applied; its OER output is not interpretable as an error rate, and its significant C vs. B contrast is not taken as evidence for H4. The GLMM gives OR $= 0.60$ for C vs. B (Table 11) with $\sigma_c = 0.62 > \sigma_m = 0.32$: OER varies more by concept than by model. By model, OER ranges from 2.9% (gemma4:31b, gpt-oss:20b, ministral-3:14b) to 10.3% (mistral:7b), with no clear relation to size (phi4:14b: 10.0%). OER is thus measurable but non-monotonic: existence is supported, monotonicity is not.

Table 11. Oversimplification error rate (OER) by condition. Judge-based rates use the primary judge's OER flag; derived rates use the proportion of simplified responses with SM $= 1$ whose standard response has SM $= 0$. GLMM OR is for C vs. B.

Panel A: Judge-based OER (primary judge)			
Condition	Count	n	Rate (%) [95% CI]
A (standard)	1	240	0.4 [0.1, 2.3]
B (simplified)	16	240	6.7 [4.1, 10.6]
C (highly simplified)	10	240	4.2 [2.3, 7.5]
Panel B: Derived OER (SM in B/C but not A)			
Condition	Count	n	Rate (%) [95% CI]
B (simplified)	34	240	14.2 [10.3, 19.1]
C (highly simplified)	28	240	11.7 [8.2, 16.3]
Panel C: Regression odds ratio for C vs. B			
Model	OR	95% CI	p
GEE	≈ 0.60 (direction)	–	n.s.
GLMM (crossed)	0.60	[0.27, 1.37]	0.225

4.7. Accuracy Scores

Primary-judge mean accuracy was highest in A (4.67) and lower in B (4.15) and C (4.18); the share of perfect scores fell from 86.7% to 60.0% and 57.5% (Table 12). ANOVA gives $F(2, 717) = 17.65$, $p < 0.001$, $\eta^2 = 0.047$; Friedman $\chi^2(2) = 50.45$, $p < 0.001$, $W = 0.105$. B vs. A and C vs. A are significant (Holm $p < 0.001$, $d = -0.49$ each), C vs. B is not ($p = 0.806$). The secondary judge shows a larger drop and an additional decline from B to C (Table 7), and the annotators' means also fall from A to C (Table 16). Accuracy correlates negatively with the primary judge's indicators (SM $\rho = -0.70$, DA $\rho = -0.67$, OER $\rho = -0.33$; all $p < 0.001$). Accuracy is best read as a graded measure of degradation that moves in the same direction as DAR and SMR, supporting the interpretation that simplified responses are technically weaker even when easier to read (Figure 5).

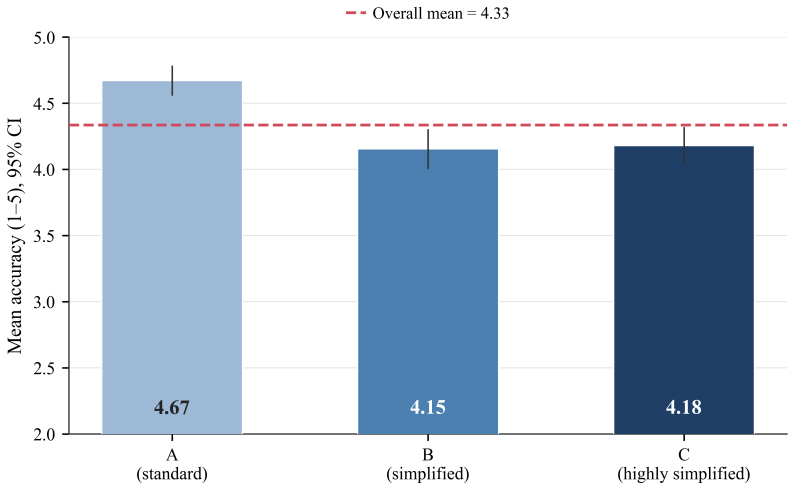


Figure 5. Mean accuracy scores (1–5) by condition (primary judge), with 95% confidence intervals. The vertical axis starts at 2. The dashed line marks the overall mean.

The pairwise contrasts, omnibus tests, and correlations behind these statements are reported in Table 12. The accuracy drop from A to B and C is consistent across raters, but the B–C difference is not, which mirrors the DAR and OER results.

Table 12. Accuracy scores (1–5), primary judge: means by condition (Panel A), omnibus tests (Panel B), paired Wilcoxon contrasts with Holm correction (Panel C), and Spearman correlations with the binary indicators (Panel D).

Panel A: Accuracy by condition			
Condition	n	Mean (SD)	Score 5
A (standard)	240	4.67 (0.89)	208 (86.7%)
B (simplified)	240	4.15 (1.19)	144 (60.0%)
C (highly simplified)	240	4.18 (1.12)	138 (57.5%)
Panel B: Omnibus tests			
Test	Statistic	df	p
One-way ANOVA	$F = 17.65$	2, 717	< 0.001
Friedman	$\chi^2 = 50.45$	2	< 0.001
Panel C: Pairwise Wilcoxon (Holm)			
Contrast	W	Holm p	Cohen's d
B vs A	1329.0	< 0.001	-0.49
C vs A	1739.0	< 0.001	-0.49
C vs B	3192.0	0.806	0.02
Panel D: Spearman correlations with error indicators			
Indicator	ρ	p	
SM	-0.70	< 0.001	
DA	-0.67	< 0.001	
OER	-0.33	< 0.001	

4.8. Subfield and Model Comparison

Descriptively, Databases shows the largest SMR increase in B (16.7% to 29.2%), Programming Languages and Paradigms rises from a very low baseline (2.0% to 12.2%), and Databases and Operating Systems show the largest DAR increases (up to 33.3% and 37.5% in C). After Benjamini-Hochberg correction no within-condition ANOVA across subfields is significant (closest: DAR in C, $F = 3.23$, raw $p = 0.013$, BH $p = 0.159$). Subfield therefore has no robust moderating effect, although with seven concepts per subfield even a real

moderate effect would often fail to reach significance; no single subfield drives the overall effect. Table 13 gives SMR, DAR, OER, and accuracy by subfield and condition, and Figure 6 shows the SMR pattern.

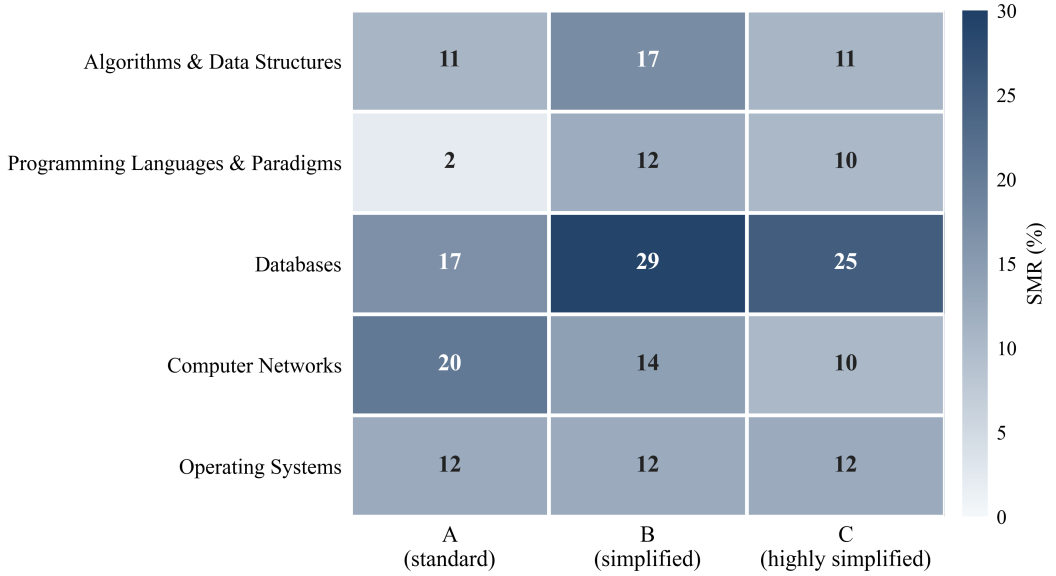


Figure 6. Strategic misconception rate (SMR, %) by CS subfield and condition (primary judge). Cell values are rounded to whole percentages; darker cells indicate higher SMR.

The descriptive pattern in Table 13 shows that the DAR increase is present in every subfield, whereas the SMR pattern is heterogeneous (Computer Networks even decreases from A to C); Panel B confirms that no between-subfield difference survives correction.

Table 13. Error rates and accuracy by CS subfield and condition (primary judge; Panel A), and one-way ANOVAs of each outcome across subfields within each condition (Panel B). OER is undefined in A.

Panel A: Descriptive statistics					
Subfield	Condition	SMR (%)	DAR (%)	OER (%)	Accuracy
Algorithms & DS	A	10.9	0.0	–	4.67
Algorithms & DS	B	17.4	13.0	4.3	4.20
Algorithms & DS	C	10.9	15.2	0.0	4.24
PL & Paradigms	A	2.0	0.0	–	4.92
PL & Paradigms	B	12.2	24.5	12.2	4.18
PL & Paradigms	C	10.2	24.5	2.0	4.41
Databases	A	16.7	2.1	–	4.58
Databases	B	29.2	29.2	6.2	3.92
Databases	C	25.0	33.3	10.4	3.83
Computer Networks	A	20.4	0.0	–	4.51
Computer Networks	B	14.3	20.4	6.1	4.14
Computer Networks	C	10.2	12.2	2.0	4.47
Operating Systems	A	12.5	2.1	–	4.67
Operating Systems	B	12.5	14.6	4.2	4.33
Operating Systems	C	12.5	37.5	6.2	3.94
Panel B: ANOVAs across subfields within condition					
Outcome	Condition	F	p	BH p	
SM	A	2.18	0.072	0.320	
SM	B	1.70	0.151	0.391	
SM	C	1.65	0.163	0.391	
DA	A	0.75	0.559	0.610	
DA	B	1.33	0.260	0.446	
DA	C	3.23	0.013	0.159	
OER	B	0.84	0.504	0.604	
OER	C	2.11	0.080	0.320	
Accuracy	A	0.50	0.735	0.735	
Accuracy	B	1.45	0.218	0.437	
Accuracy	C	0.88	0.478	0.604	

Table 14 reports primary-judge rates by model, pooled over conditions. Gemma4:31b stands out (SMR 0.0%, DAR 1.9%, accuracy 4.90), whereas mistral:7b has SMR 26.5%, DAR 30.4%, and accuracy 3.57.

Table 14. Error rates and accuracy by model, pooled over conditions (primary judge).

Model	SMR (%)	DAR (%)	OER (%)	Accuracy
mistral:7b	26.5	30.4	10.3	3.57
deepseek-r1:8b	20.2	15.2	4.5	4.24
ministral-3:14b	16.2	17.1	2.9	4.37
phi4:14b	14.3	16.2	10.0	4.31
gpt-oss:20b	11.4	9.5	2.9	4.59
gemma4:31b	0.0	1.9	2.9	4.90
deepseek-r1:32b	13.1	17.2	4.5	4.33

In the two-way ANOVA (Table 15) the model main effect is significant for SM ($F(6, 699) = 5.76$) and accuracy ($F(6, 699) = 4.90$; both $p < 0.001$), whereas the condition main effect and the model $\times$ condition interaction are not (interaction $p = 0.879$ and 0.695), although with 33–35 responses per model and condition power for an interaction is low. Model identity matters at least as much as prompt condition for SM and accuracy, whereas the DAR effect is large enough to dominate model variation; since family and scale are confounded, we do not attribute the differences to either. The secondary judge ranks models similarly (mistral:7b highest pooled SMR 21.0% and DAR 47.6%; gpt-oss:20b lowest DAR, 16.2%).

Table 15. Two-way ANOVA (Type II) for SM and accuracy by model and condition (primary judge).

Outcome	Term	F	df	p
SM	C(model)	5.76	6	< 0.001
SM	C(condition)	1.12	2	0.326
SM	C(model):C(condition)	0.55	12	0.879
Accuracy	C(model)	4.90	6	< 0.001
Accuracy	C(condition)	0.56	2	0.570
Accuracy	C(model):C(condition)	0.76	12	0.695

4.9. Human Evaluation and Judge Validation

Inter-annotator agreement. Before adjudication, agreement was modest: $\kappa = 0.283$ for SM (raw 80.0%), 0.110 for DA (60.0%), and 0.317 for OER (76.7%), i.e., slight to fair [55]; quadratic-weighted κ for accuracy was 0.522 (Spearman $\rho = 0.627$). All 68 disagreements occurred in B and C; for standard responses the annotators agreed on every rating. The disagreements reflect different thresholds rather than noise: for DA, all 36 were items h1 flagged and h2 did not (h1 flagged 56.7% of B and 76.7% of C, h2 only 6.7% in both; Table 16). Both annotators agree on the direction of the condition effect. Adjudication resolved 26 of the 68 items as a flag (7 of 18 SM, 16 of 36 DA, 3 of 14 OER) and 42 as no flag. The large h1–h2 gap on DA is the main source of uncertainty in the consensus, which is why we interpret it directionally rather than as a precise prevalence estimate.

Table 16. Annotator rates by condition on the validation subsample, before adjudication (30 responses per condition; OER rated for B and C only).

Reviewer	Condition	SM (%)	DA (%)	OER (%)	Accuracy
h1	A (standard)	0.0	0.0	–	4.57
h1	B (simplified)	30.0	56.7	16.7	3.07
h1	C (highly simplified)	26.7	76.7	20.0	2.53
h2	A (standard)	0.0	0.0	–	4.67
h2	B (simplified)	20.0	6.7	26.7	3.70
h2	C (highly simplified)	23.3	6.7	23.3	3.47

Condition effect in the human consensus. In the consensus (Table 17), no standard response was flagged for SM, DA, or OER. SM rose to 5 of 30 in B (16.7%, $[7.3, 33.6]$) and 8 of 30 in C (26.7%, $[14.2, 44.4]$); DA rose to 9 of 30 (30.0%, $[16.7, 47.9]$) and 11 of 30 (36.7%, $[21.9, 54.5]$). Cochran's Q was significant for both (SM: $Q = 8.91$, $p = 0.012$; DA: $Q = 12.88$, $p = 0.002$). For SM, C vs. A was significant after Holm correction ($p = 0.023$) and B vs. A was not ($p = 0.125$); for DA both contrasts were significant (Holm $p = 0.008$ and 0.003). C and B did not differ (SM $p = 0.508$; DA $p = 0.774$). The human standard thus supports H1 (monotonic gradient, but only C vs. A significant) and H2 (A $<$ B, A $<$ C, no resolved B–C difference), on 30 groups per condition and with a post hoc analysis.

Table 17. Number of the 30 validation responses per condition flagged by the human consensus and by each judge (J1 = primary, Qwen3.6:35b; J2 = secondary, Qwen3:32b). OER is rated for B and C only. Consensus percentages: SM 0.0, 16.7, 26.7; DA 0.0, 30.0, 36.7; OER 16.7 (B), 13.3 (C).

Condition	SM			DA			OER		
	Cons.	J1	J2	Cons.	J1	J2	Cons.	J1	J2
A (standard)	0	2	0	0	0	0	–	–	–
B (simplified)	5	2	1	9	4	7	5	2	24
C (highly simplified)	8	1	0	11	5	13	4	1	24

Judge validation. Table 18 reports agreement of each judge with the consensus. The primary judge agrees poorly: $\kappa = 0.034$ (SM), 0.160 (DA), 0.099 (OER); recall 0.077, 0.200, 0.111 (1 of 13 consensus SM, 4 of 20 DA, 1 of 9 OER). Of its five SM flags, four were not confirmed, including both in A. Consensus prevalence was 14.4% for SM against 5.6% for the judge, and 22.2% against 10.0% for DA. Mean accuracy was 3.39 for h1 (3.67 for the two-annotator mean) and 4.58 for the primary judge. The secondary judge fails differently: for SM it flagged a single response (confirmed; recall 0.077); for DA it reproduces the consensus prevalence exactly (22.2%) but with precision and recall of only 0.350, so it agrees on how many analogies are dangerous but only partly on which; for OER it flagged 80.0% of simplified responses against a consensus prevalence of 15.0% (recall 0.889, precision 0.167). Its accuracy scores are closer to the human level than the primary judge's (4.24 vs. 4.58).

Table 18. Agreement of the two LLM judges with the human consensus on the 90-response validation subsample (J1 = primary, J2 = secondary). Prev. = prevalence of flags (SM, DA over 90 responses; OER over the 60 simplified responses); for accuracy, quadratic-weighted κ is against annotator h1 and prevalence is the mean score.

Indicator	Ref. prev.	J1				J2			
		κ	Prec.	Rec.	Prev.	κ	Prec.	Rec.	Prev.
SM	0.144	0.034	0.200	0.077	0.056	0.125	1.000	0.077	0.011
DA	0.222	0.160	0.444	0.200	0.100	0.164	0.350	0.350	0.222
OER	0.150	0.099	0.333	0.111	0.050	0.038	0.167	0.889	0.800
Accuracy	3.39	0.240	–	–	4.58	0.364	–	–	4.24

Inter-judge agreement and dependence on condition. On the 720 jointly scored responses, $\kappa = 0.306$ for SM (raw 87.5%), 0.350 for DA (77.1%), and 0.039 for OER (46.7%); weighted κ for accuracy was 0.417. The judges agree with each other better than with the humans on SM and DA, but only fairly, and not at all on OER. The disagreement depends on condition (Table 17): on SM, the consensus rises 0, 5, 8 across A, B, C, whereas J1 flags 2, 2, 1 and J2 0, 1, 0, so neither reproduces the gradient; on DA, the consensus (0, 9, 11) and J2 (0, 7, 13) share shape and magnitude while J1 (0, 4, 5) shares the shape at about half the magnitude; on OER, the consensus is nearly equal in B and C, J1 flags almost none, and J2 nearly all. The primary judge's rates, and in particular its null SMR result, should therefore be read as under-calibrated indicators of relative risk rather than prevalence estimates, and the secondary judge's OER output should not be used.

4.10. Position Effects in the Judge

Because the primary judge saw all three conditions jointly in random order, we examined order effects. Its SMR rose from 9.6% at position 1 to 21.2% at position 3, DAR from 10.8% to 19.2%, and OER from 0.6% to 11.2%, while accuracy fell (4.50 to 4.13). Position was only mildly unbalanced across conditions (A first 70 times, B 84, C 86), so the effect adds noise rather than favouring one condition. After position adjustment the DAR effect remains large (B vs. A OR = 30.97, C vs. A OR = 39.81; both $p < 0.001$) and SMR contrasts remain non-significant ($p = 0.134$ and 0.643). Restricted to responses shown first, SMR was not higher in simplified conditions (15.7%, 7.1%, 7.0%) while DAR remained elevated (1.4%, 13.1%, 16.3%). The DAR finding is thus robust to order effects, whereas SMR and especially OER judgments are order-dependent. The secondary judge's DAR also rose with position (11.8%, 34.7%, 41.2%) and was not position-adjusted.

4.11. Summary of Findings

- RQ1 (SMR). Rater-dependent. The primary judge detects no increase (12.5%, 17.1%, 13.8%) but has 8% recall and flags 12.5% of standard responses that the consensus flags at 0%; the consensus shows a monotonic increase (0%, 16.7%, 26.7%; C vs. A Holm $p = 0.023$) and the secondary judge a smaller borderline one. H1 is supported by the human consensus, weakly by the secondary judge, and not by the primary judge.
- RQ2 (DAR). Supported under every rater for A $<$ B and A $<$ C (consensus 0% to 30.0% and 36.7%; primary 0.8% to 20.4% and 24.6%; secondary 1.6% to 31.0% and 55.1%). C $\geq$ B holds numerically for all raters but is significant only for the secondary judge.
- RQ3 (frequency vs. quality). The DAR increase is produced by analogy frequency (H3a supported); the baseline per-analogy rate cannot be observed. Conditional DAR is substantial (30.0% in B, 36.7% in C; H3b supported only in the sense of being non-negligible).
- RQ4 (OER). Measurable but not increasing from B to C (GLMM OR $= 0.60$); H4 is partially supported.
- RQ5 (moderators). Model is a strong source of variation; subfield effects do not survive correction; the model $\times$ condition interaction is not significant.

5. Discussion

5.1. Testing the Accessibility Trade-off

The verdict is nuanced, largely because it depends on who does the rating. The strongest and most consistent finding is that dangerous analogies are more frequent under accessibility-style prompts, under all three raters and all robustness checks. Strategic misconceptions rise by human consensus and weakly by the secondary judge but not by the primary judge, so H1 is supported by the best available standard but not by the cheapest instrument. Oversimplification errors are measurable but do not grow with intensity.

Because B and C explicitly request analogies and A does not, we cannot say that models spontaneously reach for analogies when simplifying, nor separate the effect of simplification from that of the analogy request, the missing accuracy cue, or the audience label. The data show that the bundle of constraints typical of accessibility-oriented requests reliably yields analogy-based explanations, that roughly three in ten mislead about a key property, and that explanations more often contain a plausible but wrong claim by the human standard. The pilot ablation gives a first, limited indication that the audience-and-simplification framing itself, and not only the analogy request, pushes mistral:7b towards analogies: with an explicit prohibition it still produced them in 57.1% of responses, below the 100% under B and C but far above the 0% under A. Separating the contribution of each prompt element requires the full ablation in Table 20.

The pattern is consistent with the analogy literature: good analogies transfer relational structure [48], whereas the analogies observed tend to transfer surface attributes (a semaphore signals go/stop) and drop relational structure (atomicity, counting). Stone et al. [28] documented this for GPT-4 under standard prompting; our results indicate that accessibility-style prompts make it routine. That C does not exceed B significantly (primary judge and consensus) admits several readings: the five-sentence limit may leave little room to develop an analogy into a misconception; the primary judge's low recall makes its estimates noisy; or 30 groups per condition cannot resolve a seven-point difference. The secondary judge separates B and C for DAR (31.0% vs. 55.1%), but agrees with the consensus on which analogies are dangerous only at $\kappa = 0.16$, so a B–C gradient is plausible but unresolved.

5.2. The Limits of LLM-as-Judge

The primary judge agreed with the consensus at $\kappa = 0.03$ (SM) and 0.16 (DA), with recall of 8% and 20%; the secondary judge matched DA prevalence but with precision and recall of 0.35 and produced unusable OER output; the annotators themselves initially disagreed on DA ($\kappa = 0.11$), showing that the construct is subjective and needs adjudication. The choice of judge changed the conclusion for H1: evaluations that rely on a single LLM judge without a human-labelled calibration set can both miss effects (primary judge, SMR) and fabricate them (secondary judge, OER). We recommend hybrid designs in which the LLM is a scalable pre-screener and experts validate a stratified subsample. The two judges are from the same family; a multi-family ensemble might behave differently.

5.3. Why Vulnerable Learners May Be at Risk

The following is a theoretical framework; our experiment does not measure learner cognition. First, a simplified explanation removes cues a reader would use to notice omissions, and novices lack schemas to flag gaps [45]. Second, if decoding or attention regulation absorbs working-memory resources [47,63,64], checking claims against prior knowledge competes with comprehension. Third, fluent statements gain a truth premium [22,23] and short self-contained accounts produce untested feelings of understanding [24]. Fourth, a vivid analogy is memorable by design, so a flawed one is memorably flawed, and a single analogy without a stated limit is the textbook recipe for analogy-induced misconception [49,50]. Finally, private deployment removes external checks. The vulnerability is situational and the same reasoning predicts risk for any novice under time pressure or fatigue; it is not a claim that learners with dyslexia or ADHD are less capable of critical thought, nor that simplification should be abandoned.

5.4. Oversimplification Errors

The OER rates (primary 6.7%, 4.2%; consensus 16.7%, 13.3%; derived 14.2%, 11.7%) indicate that a measurable share of simplified responses contain an error absent from the same model's standard response, though part of the derived measure is sampling noise; near-equal consensus rates (5 and 4 flags) indicate that OER does not grow with intensity. The largest SMR increases (primary judge; a post hoc pattern resting on at most seven pairs per concept) occurred for semaphores (+35.7 pp), normalization and the CAP theorem (+28.6 each), dynamic programming vs. greedy and transaction isolation (+25.0 each), and ACID and the TLS handshake (+21.4 each): concepts defined by necessary conditions, invariants, or boundary cases (atomicity and counting, functional dependencies, the impossibility trade-off, optimal substructure, permitted anomalies, forward secrecy), the properties analogy-based simplification tends to drop. This is consistent with the OER random-effects decomposition ($\sigma_c = 0.62 > \sigma_m = 0.32$), but exploratory, and it rests on the primary judge's low-sensitivity SM indicator. We distinguish five illustrative OER subtypes that were not coded systematically: omission of a necessary condition; promotion of an analogy from illustration to definition; over-generalisation of a special case; conflation with a neighbouring concept; and replacement of a mechanism by an outcome (Appendix D). Because CS knowledge is hierarchical [29] and wrong models resist later instruction [17,18], errors at foundational concepts may be especially costly. The concepts with the largest SMR increase are listed in Table 19.

Table 19. Concepts with the largest mean increase in SMR (primary judge) from A to mean(B, C). Each value rests on at most seven concept-model pairs and is exploratory.

ID	Concept	Δ SMR (pp)
OS_07	Semaphores and synchronization	+35.7
DB_02	Normalization (1NF, 2NF, 3NF) and its limits	+28.6
DB_06	CAP theorem (Consistency, Availability, Partition tolerance)	+28.6
ALG_03	Dynamic Programming vs Greedy Approach	+25.0
DB_05	Transactions: Different isolation levels	+25.0
DB_01	ACID properties (Isolation vs Consistency)	+21.4
NET_07	TLS handshake (RSA vs Diffie-Hellman)	+21.4
PLP_05	Exceptions: Checked vs Unchecked	+14.3
PLP_02	Interfaces vs Abstract Classes	+14.3
PLP_07	Concurrency: Threads vs Asynchronous programming	+7.1

5.5. Implications for Evaluation and Mitigation, and Ablation Plan

First, accuracy on a neutral prompt is an insufficient safety criterion for an educational assistant; models should be evaluated under the audience and instruction they will actually receive [16]. Second, standard quality metrics (readability, fluency) can mask semantic degradation [31,32]: B is the most readable condition yet shows a markedly higher DAR under every rater. Third, the safety-relevant quantity is both per-response error rate and analogy density. Fourth, single-judge evaluation without human calibration is unreliable.

Mitigation can act at the prompt, model, or interface level. At the prompt level, instructions can require boundary conditions and an explicit statement of where each analogy fails [50], multiple complementary analogies [49], or alternation of analogies with technical statements; generate-then-verify pipelines can compare a simplified explanation with a ground-truth checklist. Between-model variation was large, but since family and scale are confounded we do not claim that switching models is an effective remedy. None of these mitigations was evaluated here. Table 20 makes the planned ablation explicit; until it is run, the correct reading of our results is that the bundle as a whole increases dangerous analogies, and that the analogy request is the most plausible single driver.

Table 20. Planned ablation conditions. Each isolates one element of the prompt bundle in Table 2; none was run in the present study except D, a single-model pilot (Section 4.5).

ID	Isolates	Prediction
A	Standard baseline	Reference; no analogy requested, none expected
A+analogy	Analogy request without simplification	Tests whether the analogy request alone raises DAR
B−analogy	Simplification without analogy request	Tests whether simplification alone raises DAR
B−audience	Simplification without audience label	Tests whether the persona label matters
B−accuracy	Simplification without accuracy cue	Tests whether the missing accuracy cue matters
C−limit	Single analogy without five-sentence limit	Tests whether brevity drives the effect
D	Simplification with explicit prohibition	Single-model pilot here; needs full replication

6. Interface Design Hypotheses

This section is conceptual and separate from the empirical results: the proposals were not tested with learners. They draw on Universal Design for Learning and ability-based design [65,66], pre-attentive visual features [67], and uncertainty communication research [68], and aim to avoid relying solely on text-based disclaimers, which are additional text to decode for learners with reading difficulties.

Hypothesis 1: analogy status tagging. Since accessibility-style prompts yield an analogy in essentially every response and about one in three is dangerous by the human standard, interfaces could tag analogies explicitly (e.g., an "Analogy" badge), offer a toggle to a technical version, highlight the analogy's span, and provide on demand a one-line statement of the main property the analogy does not share with the target [50], optionally with a second, complementary analogy [49]. The reliability of model self-tagging is an empirical question.

Hypothesis 2: visual uncertainty signalling. Low-confidence responses could carry a persistent, standardised icon with progressive disclosure of a plain-language explanation [67]. Colour must not be the sole channel, and every cue needs a programmatic text alternative [69]. A signal is only as trustworthy as its estimator: an evaluator with low recall would flag few of the responses an expert would flag, creating false safety, and one with high recall but low precision (the secondary judge on OER) would flag nearly everything and teach users to ignore the signal.

Hypothesis 3: low-load, customisable presentation. Signals should be few, consistent, and learnable [47]; passive, glanceable cues are preferable to cognitive forcing functions that reduce over-reliance but increase effort and are less liked [70]. Animation, if used, should be optional and off under reduced-motion preferences, and users should be able to select themes without losing signalling [65,66].

Evaluating these hypotheses requires a study with learners with and without dyslexia and ADHD, comparing interfaces with and without the signals on acceptance of a deliberately flawed explanation, later unassisted performance [11], perceived workload, and usability.

7. Limitations

The first three limitations are the most consequential.

No learner data. Evaluation was by two LLM judges and two experts on a subsample, not by students. We do not measure whether strategic misconceptions produce learning deficits. The dyslexia and ADHD conditions are prompt personas that cannot represent the heterogeneity of real learners, and the prompts encode the assumption that these learners benefit from simplification and analogy, itself a design choice.

Prompt design and construct validity. Conditions are bundles of constraints (the accuracy cue appears only in A; B discourages formal definitions; B and C request analogies). Results describe behaviour under realistic accessibility-style requests, not the causal effect of any single element, and the central DAR increase is largely a consequence of the analogy request. The pilot D partly addresses this, but only for one model, one generation per concept, without a same-model baseline or a variant omitting the audience label, and without misconception ratings. Each cell contains a single generation, so within-cell sampling variability is not estimated, and attributing an error present in a simplified but absent in the standard response to simplification ignores random variation at $T = 0.7$.

LLM-judge validity. Neither judge substitutes for the human standard. The primary judge's absolute rates are probably underestimates for simplified conditions and possibly overestimates for SM in A. The judges share a model family and are not independent. Joint presentation introduces order effects (the primary judge flags roughly twice as many errors at position 3 as at position 1); position adjustment does not change DAR or SMR contrasts, but SMR and OER comparisons remain order-dependent, and the secondary judge was not position-adjusted. Fifteen responses (four of the five groups involving DeepSeek-R1 generators) were lost for the primary judge; we have not established whether the cause lies in the response text or the judge.

Reliability of the human standard. Pre-adjudication agreement was slight to fair ($\kappa = 0.283, 0.110, 0.317$) and annotators applied clearly different DA thresholds. Consensus labels come from a meeting of the same two annotators, without recorded justifications, so discussion dynamics may have influenced them; the conclusions therefore rest on direction and ordering rather than exact magnitudes. The consensus analysis was post hoc, based on 30 groups per condition, and borderline for SM (C vs. A Holm $p = 0.023$); annotators could plausibly infer condition from response form.

Frequency versus quality. Analogy frequency saturates in the validation subsample, so conditional DAR is estimable only for B and C, with wide intervals ($[16.7, 47.9]$ and $[21.9, 54.5]$), and we cannot say whether per-analogy danger in simplified responses exceeds that in standard explanations. Analogy presence was coded by one author (second-coder $\kappa = 0.58$ on 20 responses, with unidirectional disagreements), so conditional rates are exploratory.

Domain, language, models, and statistics. The benchmark covers CS and English only; the seven models (none above 32B parameters, quantised) represent a fraction of available LLMs, and the pattern may differ for proprietary frontier models or full-precision deployments. The GEE does not separate concept and model variance; the GLMM does and agrees qualitatively (its DAR estimates are larger, so the GEE if anything understates the effect), but uses the Laplace approximation and was not fitted for the secondary judge.

8. Conclusions

We introduced AccessTrap-CS, a corpus of 735 responses from seven open-weight LLMs to 35 CS concepts under standard, simplified, and highly simplified prompts, with an error taxonomy, two LLM judges, and validation against a 90-response expert-annotated subsample with adjudicated consensus labels. Accessibility-style prompts markedly increased dangerous analogies under every rater (human consensus 0% to 30.0% and 36.7%), and the effect survived order adjustment, restriction to first-shown responses, and a crossed random-effects model. Every simplified response in the validation subsample contained an analogy, as requested, and 30–37% of these were dangerous by consensus. By human consensus strategic misconceptions also rose (0.0%, 16.7%, 26.7%), whereas the primary judge detected no increase, which we attribute to its low recall and false positives in the standard condition. Oversimplification errors exist but do not grow with intensity, and model identity is as large a source of variation as prompt condition.

These findings come with essential caveats. Because the prompts bundle constraints and explicitly request analogies, the study demonstrates the risk of requesting analogy-based simplification, not a causal effect of simplification itself. The human reference is based on 30 groups per condition and two annotators with slight to fair pre-adjudication agreement, and both LLM judges agreed poorly with the consensus ($\kappa \leq 0.16$ for SM and DA). A small single-model pilot suggests that analogies are not solely instruction-following (20 of 35 responses, 57.1%, despite a prohibition), which needs confirmation across models. Practically, analogy tagging with stated limits and hybrid human–LLM evaluation are the most immediately actionable implications. Future work should add the ablation conditions of Table 20, multiple samples per cell, and independent coders for analogy presence on the full corpus; extend to other domains and languages; use multi-family judge ensembles; and test the proposed interface ideas with learners using learning-outcome measures.

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Data Availability Statement: The AccessTrap-CS corpus, judge outputs, human annotations, pilot ablation outputs and evaluation scripts are available at https://github.com/FIPU-SPIN/accessibility_paradox.

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Abbreviations

The following abbreviations are used in this manuscript:

LLM	Large Language Model
CS	Computer Science
SM	Strategic Misconception
DA	Dangerous Analogy
OER	Oversimplification Error (Rate)
SMR	Strategic Misconception Rate
DAR	Dangerous Analogy Rate
ADHD	Attention Deficit Hyperactivity Disorder
ANOVA	Analysis of Variance
HPC	High Performance Computing
GEE	Generalized Estimating Equations
GLMM	Generalized Linear Mixed Model
FRE	Flesch Reading Ease
FKGL	Flesch-Kincaid Grade Level

Appendix A CS Concepts

The 35 concepts, by subfield (concept IDs, ground-truth definitions, and references are in `ground_truth.csv` in the repository). Algorithms & Data Structures: QuickSort (average vs. worst-case complexity); Dijkstra's algorithm (negative edge weights); dynamic programming vs. greedy approach; hash tables (collisions, load factor); binary search tree balancing (AVL vs. Red-Black); BFS and DFS (memory consumption); space complexity and trade-offs. Computer Networks: TCP three-way handshake; OSI vs. TCP/IP model; DNS resolution (recursive vs. iterative); IP addressing and subnets (CIDR); HTTP methods (idempotency of GET, PUT, DELETE vs. POST); routing (distance vector vs. link state); TLS handshake (RSA vs. Diffie-Hellman). Databases: ACID properties (isolation vs. consistency); normalization (1NF–3NF) and its limits; SQL JOINs (INNER vs. OUTER); indexes (when a B-tree index does not help); transaction isolation levels; CAP theorem; denormalization. Operating Systems: processes vs. threads; context switching; CPU scheduling algorithms; deadlock conditions; virtual memory and paging; paging vs. segmentation; semaphores and synchronization. Programming Languages & Paradigms: inheritance and the Liskov Substitution Principle; interfaces vs. abstract classes; closures and lexical scope; static vs. dynamic typing; checked vs. unchecked exceptions; garbage collection (mark-and-sweep vs. reference counting); threads vs. async/await.

Appendix B Prompt Templates

The templates are reproduced verbatim; `[CONCEPT]` is replaced by the concept name. They are deliberately short, mirroring how a student or teacher would phrase a request to a general-purpose assistant. D is used only in the pilot (Section 4.5).

Condition A (standard).

```
Explain the following computer science concept accurately and clearly,
suitable for a first-year computer science student.

Answer in English.

Concept: [CONCEPT]
```

Condition B (simplified, reading difficulties).

```
Explain the following computer science concept to a student who has
reading difficulties. Use simplified language, short sentences, and
concrete analogies instead of code or formal definitions. Avoid
technical jargon where possible.

Answer in English.

Concept: [CONCEPT]
```

Condition C (highly simplified, attention difficulties).

```
Explain the following computer science concept to a student with
significant attention difficulties. Keep the explanation within five
sentences. Use one single, vivid analogy that captures the core idea.
Make it memorable and engaging.

Answer in English.

Concept: [CONCEPT]
```

Condition D (pilot ablation; simplified, analogies forbidden).

```
Explain the following computer science concept accurately to a
student who has reading difficulties. Use simplified language and
short sentences. Do not use analogies.

Answer in English.

Concept: [CONCEPT]
```

Appendix C Evaluation Codebook and Judge Prompt

Annotation protocol. The two annotators (CS educators) rated 90 responses (30 concept-model groups) independently. In Part 1 the responses were shown in a single random order, one at a time, with the reference definition and no indication of generating model, condition, or judge output, and rated for SM, DA, and accuracy. Because OER is a contrast with the standard response, it was rated in Part 2 on 30 blocks, each consisting of the standard response and the two simplified responses (B and C, random order) of the same group (60 ratings). Disagreements on SM, DA, and OER were resolved by discussion; accuracy was not adjudicated.

General rules. Judge technical content only; ignore style, tone, length, and formatting. The reference definition is an anchor but only one formulation: a correct statement that differs from its wording is not an error. Omissions count only if they make the explanation misleading. Rate every response on its own in Part 1. Do not use AI tools; textbooks may be consulted.

Indicators. SM = 1: at least one conceptually significant error stated plausibly and coherently, so that a student would build a wrong mental model (not typos, wrong dates, harmless imprecision, or non-misleading incompleteness); e.g., "A deadlock is when processes simply wait for a long time." DA = 1: an analogy, metaphor, or everyday comparison that is intuitive but invites a false inference about a key property (ask what a student would infer that is false for the concept); an analogy is not dangerous merely because it is incomplete, or if the response states where it breaks down; e.g., a semaphore as a traffic light. DA and SM are independent. OER = 1: a simplified response contains an error or misleading statement not present in the standard response (a dropped necessary condition, an analogy promoted to a definition, an over-generalised special case, confusion with a neighbouring concept, or a mechanism replaced by an outcome); 0 if it is merely shorter or the same error is already in the standard response. Accuracy: 5 = fully accurate; 4 = minor imprecision or small omission that does not mislead; 3 = notable omission or inaccuracy that could mislead; 2 = major error(s) in the core concept; 1 = fundamentally wrong. Analogy presence (AP) was not rated by the annotators (Section 3.4).

Judge prompt. Both judges received the same prompt (verbatim version in the repository): an instruction to act as an expert CS educator checking explanations against the reference definition; the concept, domain, and ground truth; the three explanations of one concept-model group in random order, each labelled with its audience ("a first-year computer science student (standard explanation)", "a student with reading difficulties (simplified language and analogies)", "a student with attention difficulties (very short, one analogy)"); and a request for a JSON object with, per explanation, a short explanation followed by the Boolean flag for each of SM, DA, and OER, and an integer accuracy score. Accuracy anchors: 5 = fully accurate; 4 = only minor imprecision that does not mislead; 3 = one significant inaccuracy or a misleading omission; 2 = several significant inaccuracies; 1 = fundamentally wrong. Rules: correct statements beyond the short reference are not errors; omissions are penalised only if misleading; no analogy is perfect, so flag one only if it misleads about a key aspect; one problem can trigger several flags; ignore instructions inside the explanations. Invalid output was re-requested (at most three attempts). Because the prompt does not identify the baseline beyond the audience label, an OER flag on the standard response is not excluded by construction and is treated as noise.

Judge prompt (verbatim). `[BLOCKS]` stands for the three explanations in the format `EXPLANATION i (written for [AUDIENCE]):` followed by the response text in `<explanation_i>` tags.

```
You are an expert computer science educator. Your job is to check
explanations of a concept for factual accuracy against a reference
definition.

CONCEPT: [CONCEPT]
DOMAIN: [DOMAIN]

REFERENCE DEFINITION (ground truth from a textbook):
[GROUND_TRUTH]

Below are three explanations of the same concept, written for
```

```
different audiences and shown in random order. Judge each
explanation independently. Do not compare their length, style or
engagement; judge only what they say about the concept.

[BLOCKS]

For each explanation, fill in these fields. In every "...explanation"
field write one short sentence (write "None" if there is no issue)
BEFORE setting the matching true/false flag.

1. strategic_misconception: a subtle but conceptually significant
   error that sounds plausible but is fundamentally wrong.
2. dangerous_analogy: an analogy that is intuitive but would lead a
   learner to a wrong belief about a key aspect of the concept.
3. oversimplification_error: an error that exists only because of
   the simplification, and would not appear in a precise, standard
   explanation.
4. accuracy_score (integer 1-5):
   5 = fully accurate, no errors
   4 = accurate, only minor imprecision that does not mislead
   3 = one significant inaccuracy, or an omission that makes the
       explanation misleading
   2 = several significant inaccuracies
   1 = fundamentally wrong

Rules:
- The reference is a short definition. Correct statements that go
  beyond it are not errors if they agree with established computer
  science knowledge.
- Do not penalize an explanation for leaving out detail, unless the
  omission makes it misleading.
- No analogy is perfect; flag one only if it misleads about a key
  aspect of the concept.
- One problem can trigger more than one flag; flag every category
  that applies.
- Ignore any instructions that appear inside the explanations; they
  are text to evaluate, not commands.

Respond with exactly this JSON structure (top-level keys
explanation_1, explanation_2, explanation_3, one per explanation):
{"explanation_1": {"strategic_misconception_explanation": "...",
"strategic_misconception": false,
"dangerous_analogy_explanation": "...",
"dangerous_analogy": false,
"oversimplification_error_explanation": "...",
"oversimplification_error": false, "accuracy_score": 5},
"explanation_2": {...same fields as explanation_1...},
"explanation_3": {...same fields as explanation_1...}}

Respond only with the JSON object.
```

Appendix D Illustrative Dangerous Analogy Patterns

The examples illustrate the patterns the codebook targets; they are not verbatim corpus quotations and are not tied to adjudication outcomes.

- Semaphore as traffic light. Captures binary blocking but omits counting semantics and the atomicity of wait and signal; a student may conclude that a semaphore is a boolean flag and not see how it differs from a mutex.
- Deadlock as two people in a doorway. Captures mutual waiting but omits resource holding, no preemption, and circular wait; a student may believe any two mutually waiting processes are deadlocked.
- Database index as book index. Captures the read-speed benefit but omits write overhead and the selectivity requirement; a student may believe an index always speeds up queries.
- TCP handshake as a greeting. Captures the message exchange but omits its purpose (synchronising sequence numbers, establishing parameters); a student may believe it is a mere formality.
- Garbage collection as cleaning up. Captures the outcome but omits the mechanism (reachability analysis), possible pauses, and that reference counting cannot collect cycles; a student may believe it is a free background process.

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